Week-1 Module - C & C++ Programming

1. C programming syntax

C is a procedural programming language used for developing System and application software

Basic structure of a C program:

1. Header files
2. Main function
3. Variable Declaration
4. Statements
5. Return Statements

Example:

```
#include <stdio.h>
int main()
{
  printf("Helloworld")
return 0;
}
```

2. Conditional statements are used to make decisions in a program

Types:

1. If
2. If-else
3. else-if ladder
4. Switch

- If checks condition
- else executes when condition is false

Example:

```
#include<stdio.h>
int main()
{
int num = 10;
if (num >5)
{
  Printf("smaller");
}
return 0;
}
```

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DATE: __/__/__
PAGE: ____

2. Loops in C

Loops are used to execute a block of code repeatedly

Types:
1. for loop
2. while loop
3. do-while loop

Example:

```
include <stdio.h>
int main()
{
for(int i=1;i<=5;
i++)
{
printf("%d\n",i
}
return 0;
}
```

Output:
1
2
3
4
5

Square pattern

```
#include <iostream>
using namespace std;
int main ()
{
for (int i=1; i<=5;i++)
{
for (int j=1; j<=5; j++)
{
Cout <<"*";
}
Cout <<endl;
}
return 0;
}
```

DATE: __/__/____
PAGE: ________

Ouput:

```
 * * * * *
* * * * *
* * * * *
* * * * *
* * * * *
```

Hollow square

```
# include <iostream>
using name space std;
int main()
{
  for(int i = 1; i<=5; i++)
  {
    for(int j = i;j<=5; j++)
    {
      if(i==1||i==5||j==7||j==5)
        Cout << "*";
      else
        cout << " ";
    }
    Cout << end;
  }
  return 0;
}
```

DATE : / /
PAGE :

Pyyamid pattern

```
# include<iostream>
using namespace std;
int main()
{
  for(int i=1; i<=5; i++)
  {
    for(int j=1; j<=5; j++)
      cout<<" ";
   for(int k=1; k=2*i-1;k++)
    cout<<"*";
    cout endl;
  }
  return 0;
}
```

output

```
        *
      * * *
    * * * * *
 * * * * * * * *
* * * * * * * * *
```

Inverted Triangle Pattern

```
#include<iostream>
using namespace std;
int main()
{
 for(int i=5; i>=1; i--)
  {
   for(int j=1; j<=i; j++)
   {
     Cout<<"*";
   }
   Cout<<endl;
  }
   return 0;
  }
```

Output

```
*****
****
***
**
*
```

DATE: / /
PAGE:

Output:

```
* * * * *
*     *
*     *
*     *
* * * * *
```

Triangle pattern

```
# includ <iostream>
using namespace std;
int main()
{
  for(int i=1; i<=5; i++)
  {
    for(int j=1; j<=5; j++)
    {
      cout<<"*";
    }
    cout<<endl;
  }
  return 0;
}
```

Output:

```
*
* *
* * *
* * * *
* * * * *
```

Good Write

Diamond Pattern

```
# include<iostream>
using namespace std;

int main()
{
 for (int i=1; i<=5; i++)
 {
   for(int j=1; j<=5-i; j++)
     Cout<<" ";

   for (int K=1; K<=2*i-1; K++)
     Cout<<"*";
     Cout<<endl;
 }
 for(int i=4; i>=1; i--)
 {
   for (int j=1; j<=5-i; j++)
     Cout<<" "

   for(int k=1; k<=2*i-1; k++)
     Cout<< "*";
   Count<< endl;
 }
   return 0;
}
```

Output:

```
   *
  ***
 *****
*******
 *****
  ***
   *
```

Half diamond Pattern

```
# include <iostream>
using namespace std;

int main()
{
  for(int i= 1; i<=5; i++)
  {
   for(int j=1; j<=5; j++)
      Cout << "*";
   Cout<<endl;
  }
  return 0;
}
```

Output

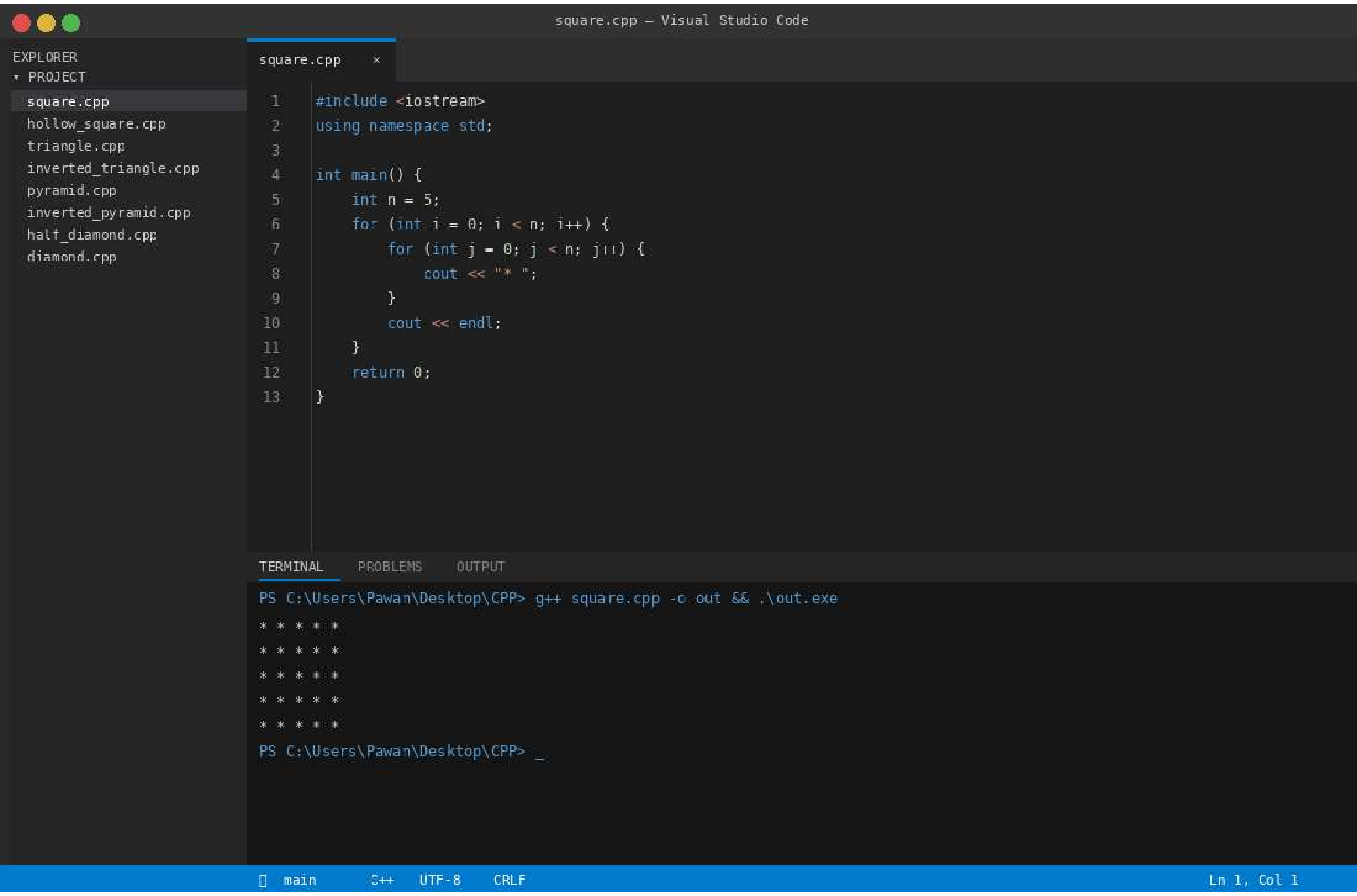
```
*
**
***
****
*****
****
***
**
*
```

Inverted pyramid pattern

```
# include<iostream>
using namespace std;
int main()
{
 for (int i=5; i>=1; i--)
 {
  for(int j=1; j<=5; j++)
      cout<<" ";

  for(int k=1; k<=2*i-1; k++)
    cout<<"*";
   cout<<endl;
  }
   return 0;
 }
```

Output:

```
*********
 *******
  *****
   ***
    *
```


square.cpp — Visual Studio Code
EXPLORER
▾ PROJECT
square.cpp
hollow_square.cpp
triangle.cpp
inverted_triangle.cpp
pyramid.cpp
inverted_pyramid.cpp
half_diamond.cpp
diamond.cpp
square.cpp ×
#include <iostream>
using namespace std;

int main() {
    int n = 5;
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            cout << "* ";
        }
        cout << endl;
    }
    return 0;
}
TERMINAL
PROBLEMS
OUTPUT
PS C:\Users\Pawan\Desktop\CPP> g++ square.cpp -o out && .\out.exe
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
PS C:\Users\Pawan\Desktop\CPP> _
main
C++
UTF-8
CRLF
Ln 1, Col 1

hollow_square.cpp — Visual Studio Code

EXPLORER

▾ PROJECT

- square.cpp
- hollow_square.cpp
- triangle.cpp
- inverted_triangle.cpp
- pyramid.cpp
- inverted_pyramid.cpp
- half_diamond.cpp
- diamond.cpp

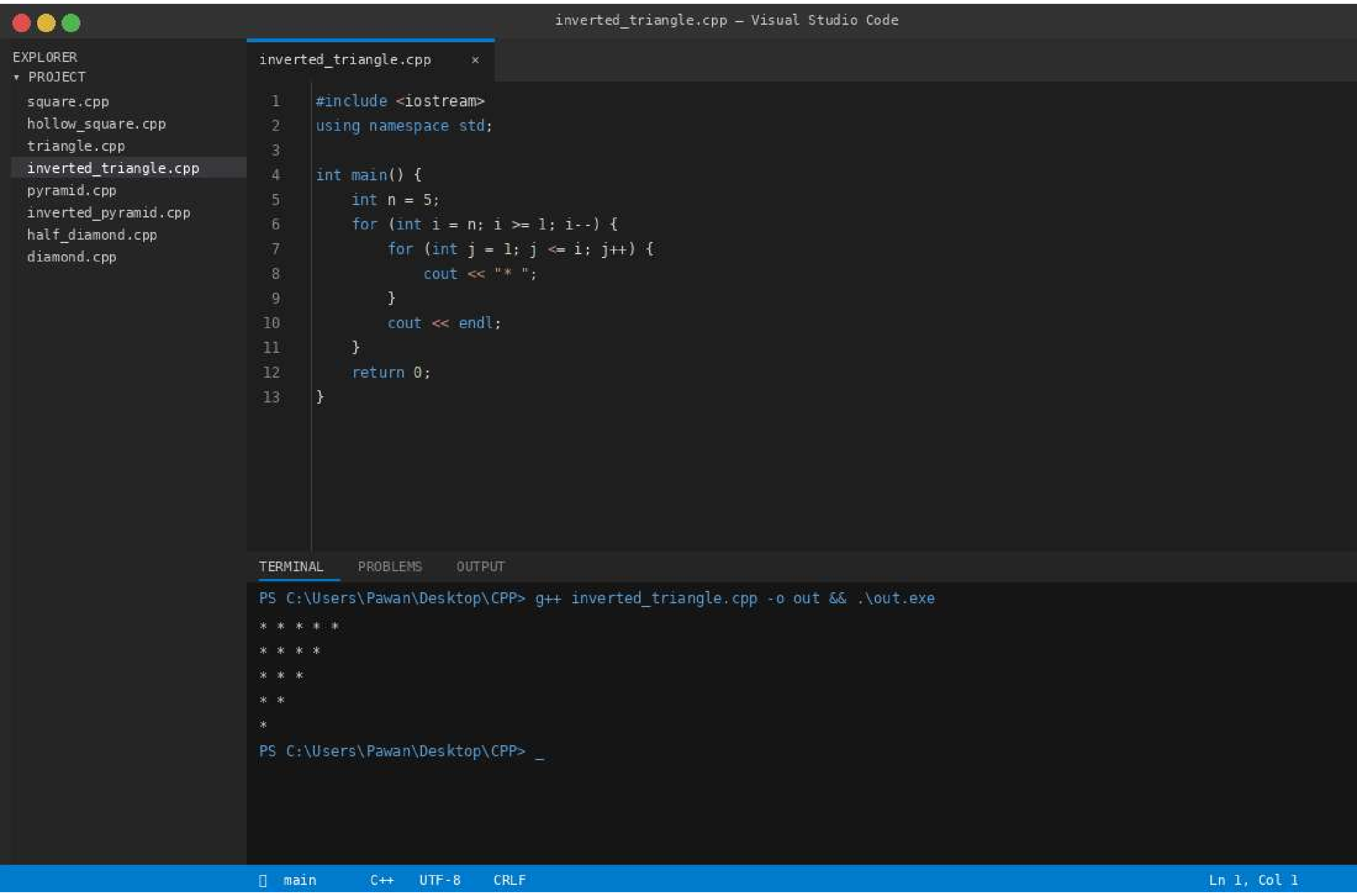
hollow_square.cpp ×

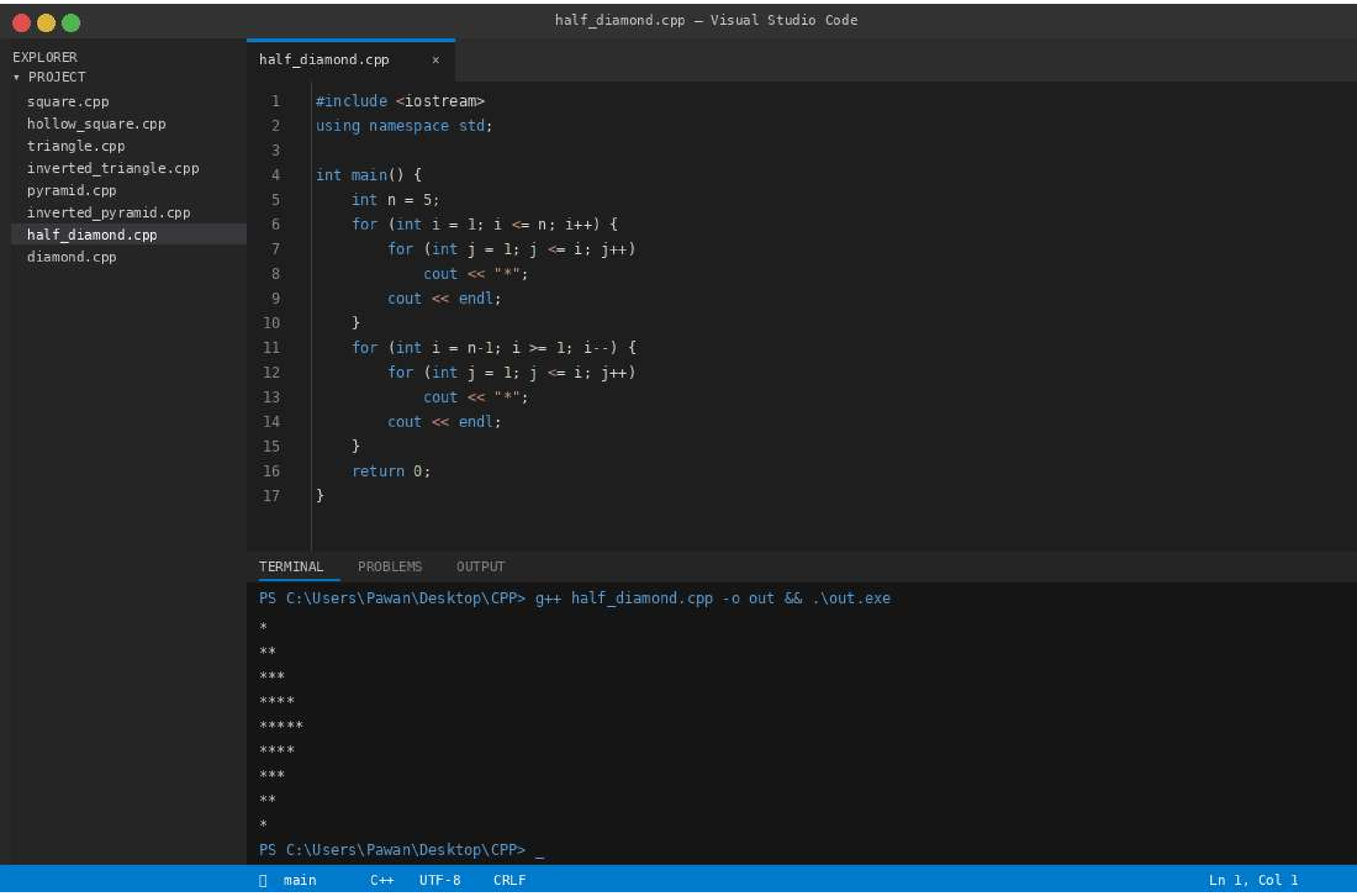
```cpp
#include <iostream>
using namespace std;

int main() {
    int n = 5;
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            if (i==0 || i==n-1 || j==0 || j==n-1)
                cout << "* ";
            else
                cout << "  ";
        }
        cout << endl;
    }
    return 0;
}
```

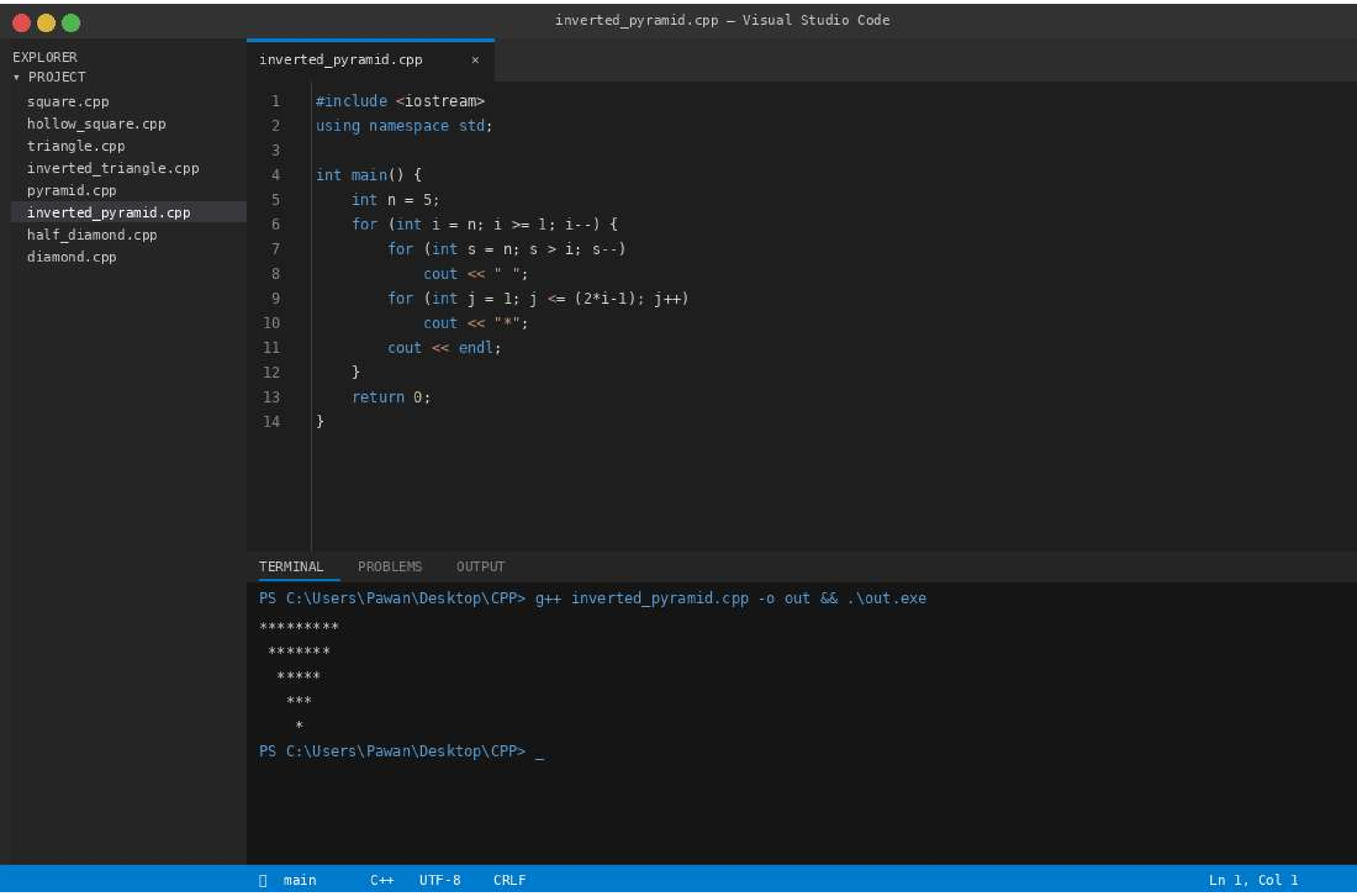
TERMINAL PROBLEMS OUTPUT

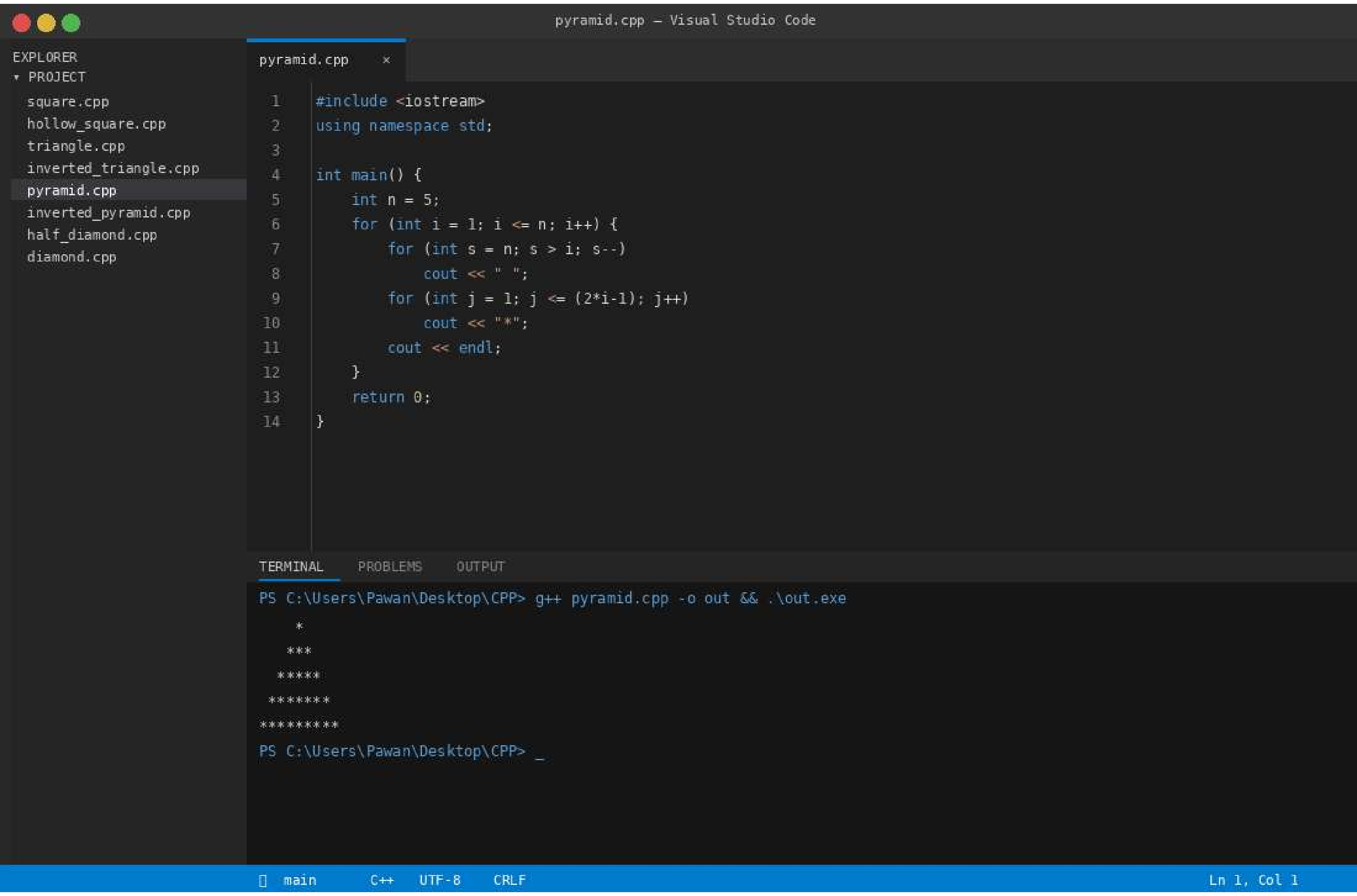
```
PS C:\Users\Pawan\Desktop\CPP> g++ hollow_square.cpp -o out && .\out.exe
* * * * *
*       *
*       *
*       *
* * * * *
PS C:\Users\Pawan\Desktop\CPP> _
```

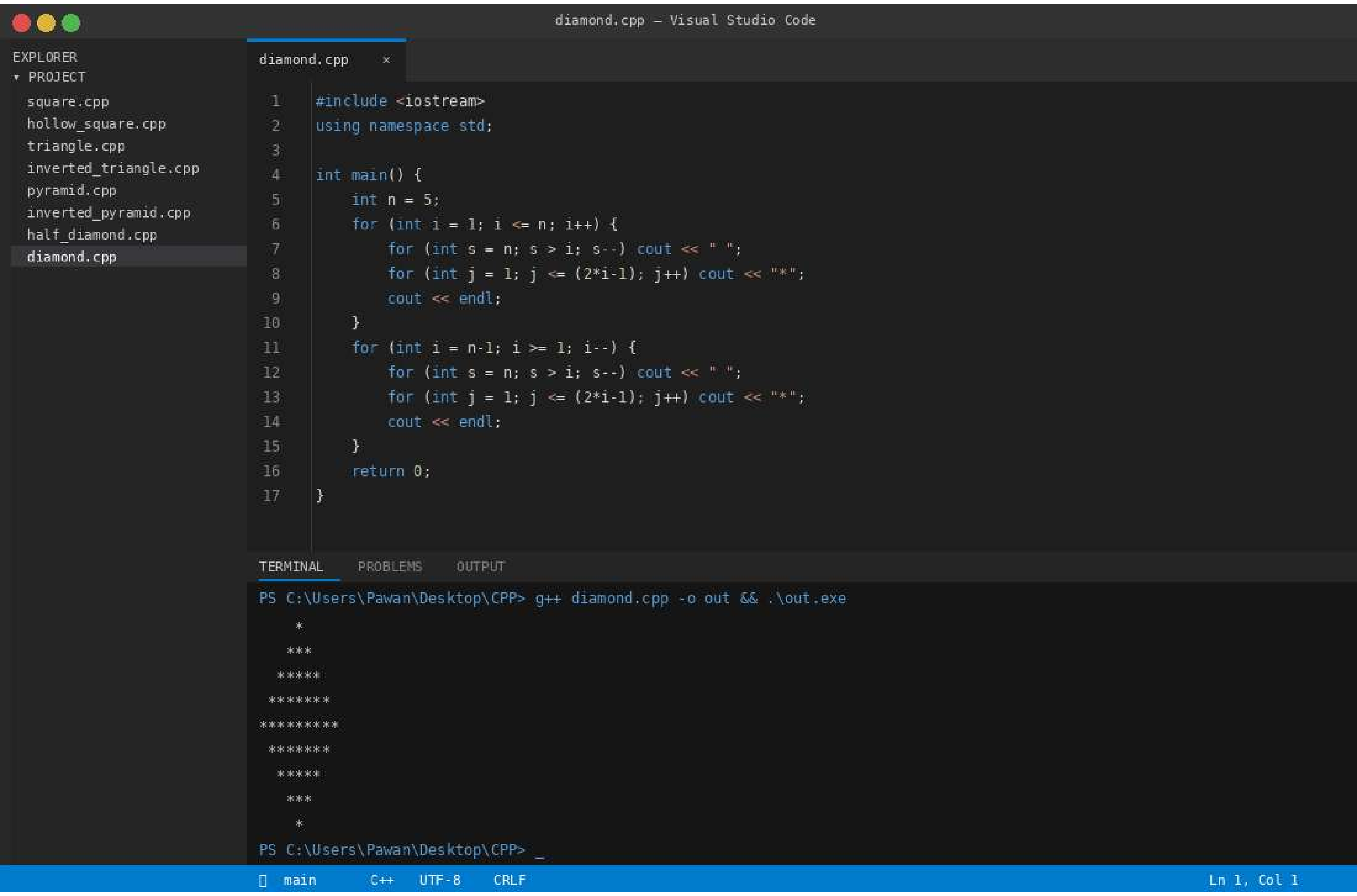
main C++ UTF-8 CRLF Ln 1, Col 1


inverted_triangle.cpp — Visual Studio Code
EXPLORER
PROJECT
square.cpp
hollow_square.cpp
triangle.cpp
inverted_triangle.cpp
pyramid.cpp
inverted_pyramid.cpp
half_diamond.cpp
diamond.cpp
inverted_triangle.cpp
#include <iostream>
using namespace std;
int main() {
int n = 5;
for (int i = n; i >= 1; i--) {
for (int j = 1; j <= i; j++) {
cout << "* ";
}
cout << endl;
}
return 0;
}
TERMINAL
PROBLEMS
OUTPUT
PS C:\Users\Pawan\Desktop\CPP> g++ inverted_triangle.cpp -o out && .\out.exe
* * * * *
* * * *
* * *
* *
*
PS C:\Users\Pawan\Desktop\CPP> _
main
C++
UTF-8
CRLF
Ln 1, Col 1

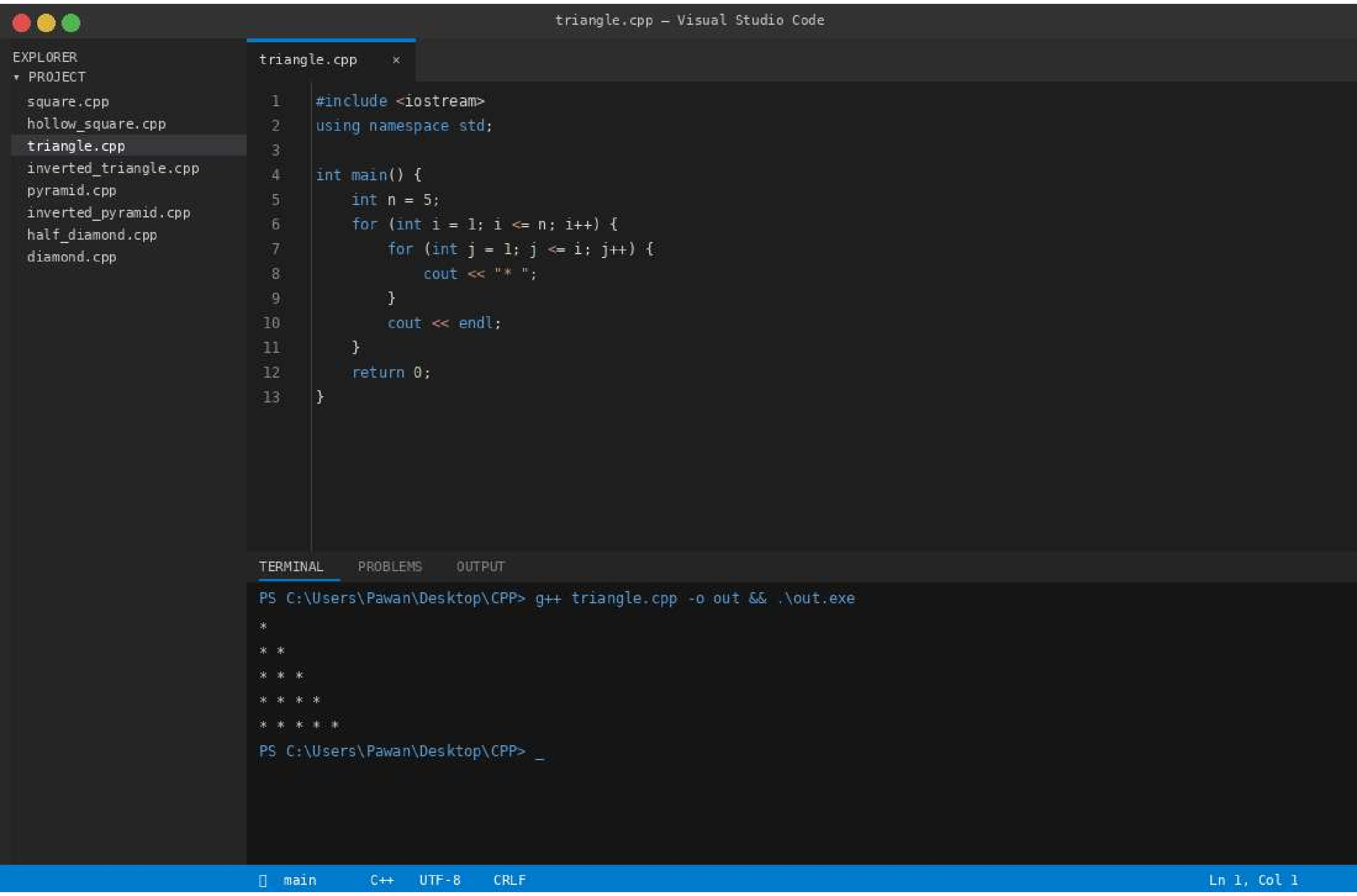

half_diamond.cpp — Visual Studio Code
EXPLORER
▾ PROJECT
square.cpp
hollow_square.cpp
triangle.cpp
inverted_triangle.cpp
pyramid.cpp
inverted_pyramid.cpp
half_diamond.cpp
diamond.cpp
half_diamond.cpp ×
#include <iostream>
using namespace std;
int main() {
int n = 5;
for (int i = 1; i <= n; i++) {
for (int j = 1; j <= i; j++)
cout << "*";
cout << endl;
}
for (int i = n-1; i >= 1; i--) {
for (int j = 1; j <= i; j++)
cout << "*";
cout << endl;
}
return 0;
}
TERMINAL
PROBLEMS
OUTPUT
PS C:\Users\Pawan\Desktop\CPP> g++ half_diamond.cpp -o out && .\out.exe
*
**
***
****
*****
****
***
**
*
PS C:\Users\Pawan\Desktop\CPP> _
main
C++
UTF-8
CRLF
Ln 1, Col 1


inverted_pyramid.cpp — Visual Studio Code
EXPLORER
▾ PROJECT
square.cpp
hollow_square.cpp
triangle.cpp
inverted_triangle.cpp
pyramid.cpp
inverted_pyramid.cpp
half_diamond.cpp
diamond.cpp
inverted_pyramid.cpp ×
#include <iostream>
using namespace std;
int main() {
int n = 5;
for (int i = n; i >= 1; i--) {
for (int s = n; s > i; s--)
cout << " ";
for (int j = 1; j <= (2*i-1); j++)
cout << "*";
cout << endl;
}
return 0;
}
TERMINAL
PROBLEMS
OUTPUT
PS C:\Users\Pawan\Desktop\CPP> g++ inverted_pyramid.cpp -o out && .\out.exe
****
****
*****
***
*
PS C:\Users\Pawan\Desktop\CPP> _
main
C++
UTF-8
CRLF
Ln 1, Col 1


pyramid.cpp — Visual Studio Code
EXPLORER
▾ PROJECT
square.cpp
hollow_square.cpp
triangle.cpp
inverted_triangle.cpp
pyramid.cpp
inverted_pyramid.cpp
half_diamond.cpp
diamond.cpp
pyramid.cpp ×
#include <iostream>
using namespace std;
int main() {
int n = 5;
for (int i = 1; i <= n; i++) {
for (int s = n; s > i; s--)
cout << " ";
for (int j = 1; j <= (2*i-1); j++)
cout << "*";
cout << endl;
}
return 0;
}
TERMINAL
PROBLEMS
OUTPUT
PS C:\Users\Pawan\Desktop\CPP> g++ pyramid.cpp -o out && .\out.exe
*
***
*****
****
****
PS C:\Users\Pawan\Desktop\CPP> _
main
C++
UTF-8
CRLF
Ln 1, Col 1


diamond.cpp — Visual Studio Code
EXPLORER
▾ PROJECT
square.cpp
hollow_square.cpp
triangle.cpp
inverted_triangle.cpp
pyramid.cpp
inverted_pyramid.cpp
half_diamond.cpp
diamond.cpp
diamond.cpp ×
#include <iostream>
using namespace std;

int main() {
    int n = 5;
    for (int i = 1; i <= n; i++) {
        for (int s = n; s > i; s--) cout << " ";
        for (int j = 1; j <= (2*i-1); j++) cout << "*";
        cout << endl;
    }
    for (int i = n-1; i >= 1; i--) {
        for (int s = n; s > i; s--) cout << " ";
        for (int j = 1; j <= (2*i-1); j++) cout << "*";
        cout << endl;
    }
    return 0;
}
TERMINAL
PROBLEMS
OUTPUT
PS C:\Users\Pawan\Desktop\CPP> g++ diamond.cpp -o out && .\out.exe
    *
   ***
  *****
 ****
****
 ****
  *****
   ***
    *
PS C:\Users\Pawan\Desktop\CPP> _
main
C++
UTF-8
CRLF
Ln 1, Col 1



triangle.cpp — Visual Studio Code
EXPLORER
▾ PROJECT
square.cpp
hollow_square.cpp
triangle.cpp
inverted_triangle.cpp
pyramid.cpp
inverted_pyramid.cpp
half_diamond.cpp
diamond.cpp
triangle.cpp ×
#include <iostream>
using namespace std;
int main() {
int n = 5;
for (int i = 1; i <= n; i++) {
for (int j = 1; j <= i; j++) {
cout << "* ";
}
cout << endl;
}
return 0;
}
TERMINAL
PROBLEMS
OUTPUT
PS C:\Users\Pawan\Desktop\CPP> g++ triangle.cpp -o out && .\out.exe
*
* *
* * *
* * * *
* * * * *
PS C:\Users\Pawan\Desktop\CPP> _
main
C++
UTF-8
CRLF
Ln 1, Col 1